

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			Indicative content X identified as hydrochloric acid Y identified as sodium chloride Z identified as ethanoic acid X is the stronger acid – more vigorous reaction / produces hydrogen more quickly; greater temperature increase / more exothermic Y does not react so cannot be an acid / must be sodium chloride Y must be sodium chloride – magnesium less reactive than sodium so can't displace it Z is the weaker acid – less vigorous reaction / produces hydrogen less quickly; smaller temperature rise / less exothermic magnesium + hydrochloric acid → magnesium chloride + hydrogen $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$ magnesium + ethanoic acid → magnesium ethanoate + hydrogen $\text{Mg} + \text{CH}_3\text{COOH} \rightarrow (\text{CH}_3\text{COO})_2\text{Mg} + \text{H}_2$ 5-6 marks All three identified; clear reasoning; good attempt at equation <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	2		4	6		4

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				3-4 marks At least two identified; some reasoning; reference to named salt and/or hydrogen as products <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks At least one identified; reference to gas/hydrogen as product <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)			<u>iron(II)</u> ions will produce a <u>green precipitate</u> (1) <u>iron(III)</u> ions will produce a <u>brown precipitate</u> (1) if no reference to precipitate award (1) for iron(II) green and iron(III) brown award (1) if correct precipitate colours given but assigned to incorrect ions	2			2		2
				Question 6 total	4	0	4	8	0	6